CSNS RCS IPM — Ion-mode scan results

Code: Virtual-IPM 2.3.1. **Reference:** NIMA **1092** (2026) 171809 (CSNS RCS IPM). **Campaign:** round-beam ion-mode matrix, **2070 / 2070** runs complete. **Species:** H_2^+ (2 u), H_2O^+ (18 u), N_2^+ (28 u) — the three ToF peaks in the paper. **Statistics:** 100000 secondaries per run; quoted σ is the RMS of **detected** x.

This report summarises the ion-mode parameter scan parallel to the e-mode replan (REPORT.md §8) and the fine C/D e-mode scan (§9). Combined e-mode + ion-mode write-up: CSNS_IPM_REPORT.md / CSNS_IPM_REPORT.pdf. Literature comparison (2006–2026): LITERATURE_REVIEW.md / LITERATURE_REVIEW.pdf. Unlike e-mode, **no dense B-scan** was run: every block uses only B_Y ∈ 0, 200, 1000 G (no guide / mid checkpoint / design 0.1 T), because closed ion scans already showed expansion **flat** vs B.

1. Scan matrix

Block	What is scanned	Beams	$\sigma_x = \sigma_Y$	B [G]	Power [kW]
A	Reference beams	inj. 80 MeV; ext. 1.6 GeV; inj. 80 MeV	25 / 10 / 10 mm	0, 200, 1000	100–500
B	Beam size	inj.; ext.	3–20 mm (1 mm)	0, 200, 1000	100–500
C	Cage voltage	inj. only	10 mm	0, 200, 1000	100, 500
D	Beam offset	inj.; ext.	10 mm	0, 200, 1000	100, 500

Common settings: round beams ($\sigma_Y = \sigma_x$); ideal cage E_Y = -V/231 mm (ions collected at y_min); circular 3-bunch train; simulation 12 μs / 8000 Boris steps; $N_b = 7.8\times10^{12}\times(P/100\text{ kW})$. SC-off reference: H_2^+ @ 100 kW only (mass-independent without bunch fields, shared).

Re-run: ./scripts/run_imode_replan.sh · Evaluate: python3 scripts/evaluate_imode.py

2. Takeaways

1. **Guiding B does not fix ion-mode profiles.** Expansion is flat between 0 and 200 G and changes only slightly at 0.1 T — opposite to e-mode, where 200–300 G restores the profile.
2. **Ion distortion scales with power and inversely with beam size.** At 0.1 T, H_2^+ injection at σ=10 mm goes from +82% (100 kW) to +422% (500 kW); at σ=3 mm, 500 kW gives obtained σ ~ 56 mm (+1700%+) for all three species.
3. **Lighter ions expand more at injection; heavier ions can be worse at extraction** (longer ToF). At 0.1 T / 100 kW / σ=10 mm: inj. H_2^+ +82%, H_2O^+ +64%, N_2^+ +56%; ext. H_2^+ +32%, H_2O^+ +55%, N_2^+ +50%.
4. **Cage voltage matters for ions (ToF / dwell time).** Expansion stays **positive** at every voltage — no e-mode-style sign flip. Higher V shortens drift and **reduces** the kick (H_2^+ inj. 10 mm @ 1000 G: +198% at 5 kV → +70% at 30 kV).
5. **Beam offset:** Δ x is translation-invariant; Δ y (toward/away from the detector) changes expansion by a few percent.
6. **Extraction H_2^+ numbers are a lower bound.** Ions are generated 125 ns before the first field-carrying bunch (generation window centred at 80 ns, tracking train at 204.6 ns). A line-charge model aligned to the bunch gives ≈ +100 % for H_2^+ at extraction (0 G, 100 kW) instead of the +33 % obtained. Injection is correctly aligned. See LITERATURE_REVIEW.md §3.5.

3. Block A — reference beams vs B

Three round reference beams at 0, 200, and 1000 G, all powers, all species.

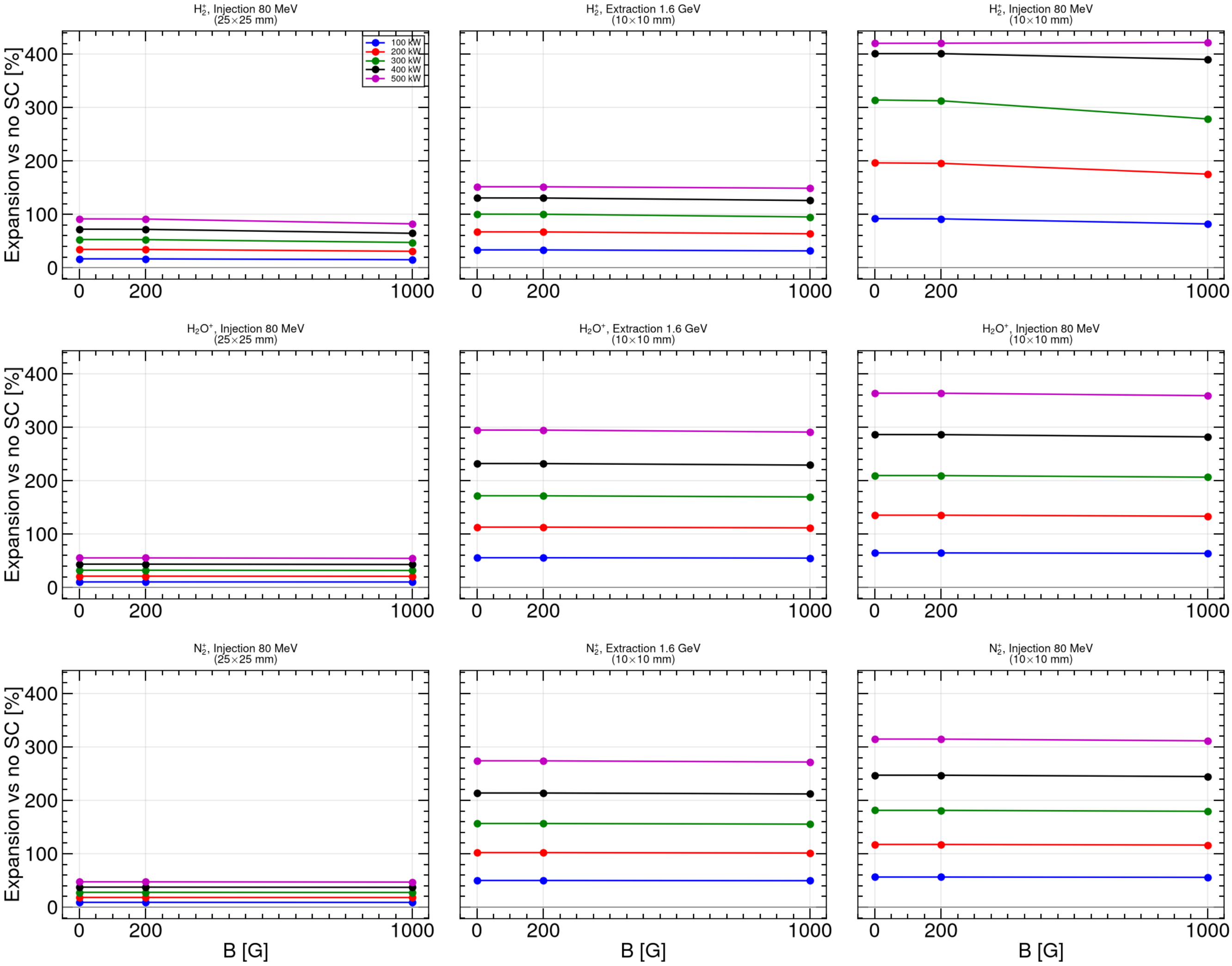

Figure 1. Profile expansion vs no-SC [%] for the three reference beams. Each panel is one species; columns are painted injection (25×25 mm), extraction (10×10 mm), and small injection (10×10 mm).

Expansion is **essentially independent of B** between 0 and 200 G (cyclotron radii are metres). The 0.1 T point is sometimes slightly lower but never restores the profile.

Table 1 — expansion at 100 kW [%]

Species	Inj. 25 mm 0 G	200 G	1000 G	Inj. 10 mm 0 G	200 G	1000 G	Ext. 10 mm 0 G	200 G	1000 G
H_2^+	+16.7	+16.7	+15.0	+92.0	+91.6	+82.1	+33.5	+33.4	+31.6
H_2O^+	+10.4	+10.4	+10.2	+64.7	+64.7	+63.8	+55.6	+55.6	+54.9
N_2^+	+8.9	+8.9	+8.9	+56.3	+56.2	+55.7	+49.9	+49.9	+49.5

Painted injection (25 mm) matches the elliptical-beam §1 numbers (+16% H_2^+). The 10 mm injection family is the most distorted at every B.

Extraction H_2^+ (+33 % at 0 G) is **under-estimated**: the generation bunch leads the tracking train by 125 ns, so H_2^+ has already drifted ≈ 40 mm when the kick arrives. H_2O^+ / N_2^+ barely move in that interval. A future extraction run should set the tracking LongitudinalOffset to −80 ns. Do not re-run the existing 100k CSVs.

4. Block B — beam-size scan (3–20 mm)

Size scan at 0, 200, and 1000 G; injection and extraction; 100–500 kW.

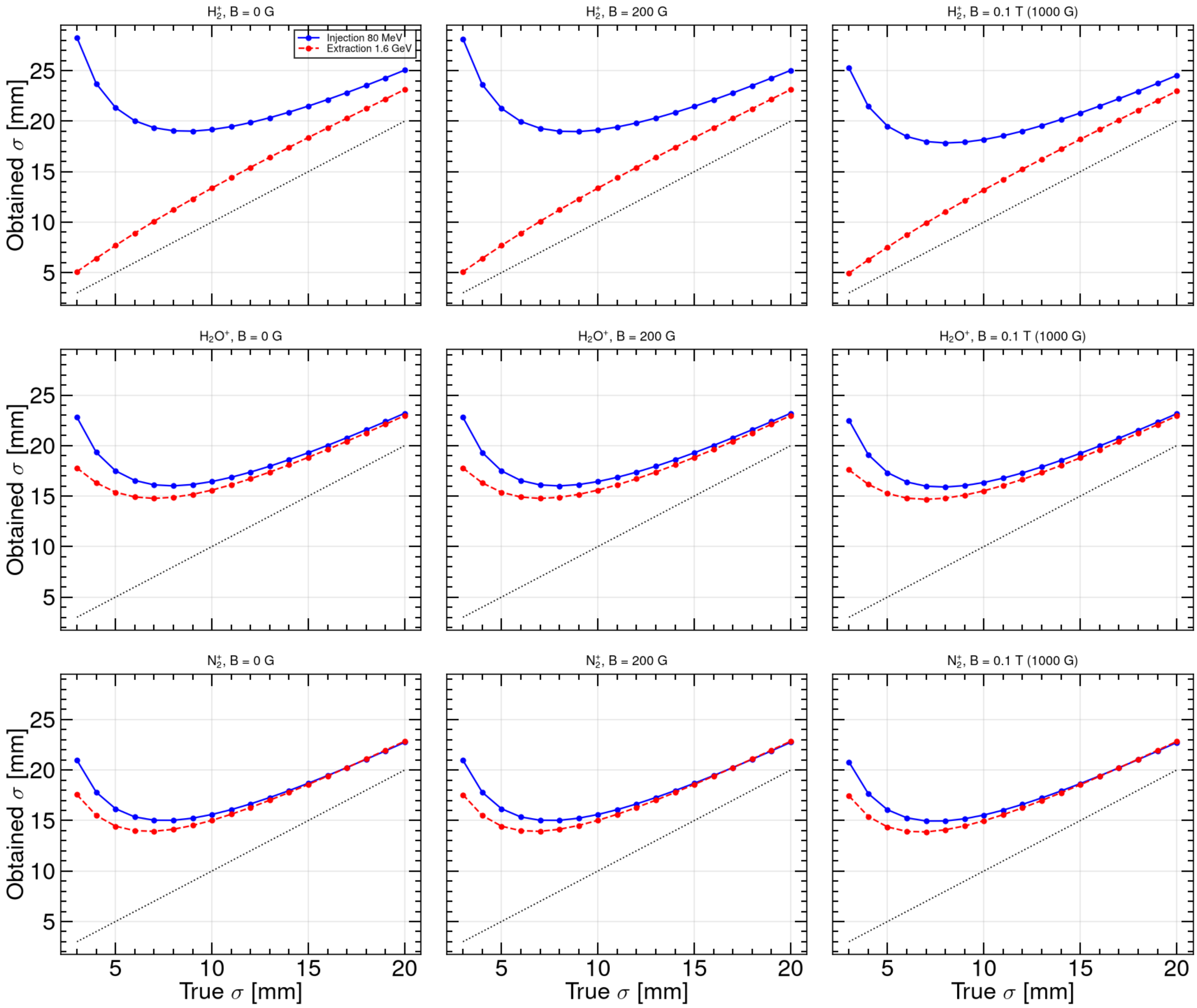

Figure 2. Obtained σₓ vs true σₓ at **0.1 T**, 100 kW. Dashed line: obtained = true. Every point lies **above** the diagonal at all three B fields — space charge always inflates the detected width.

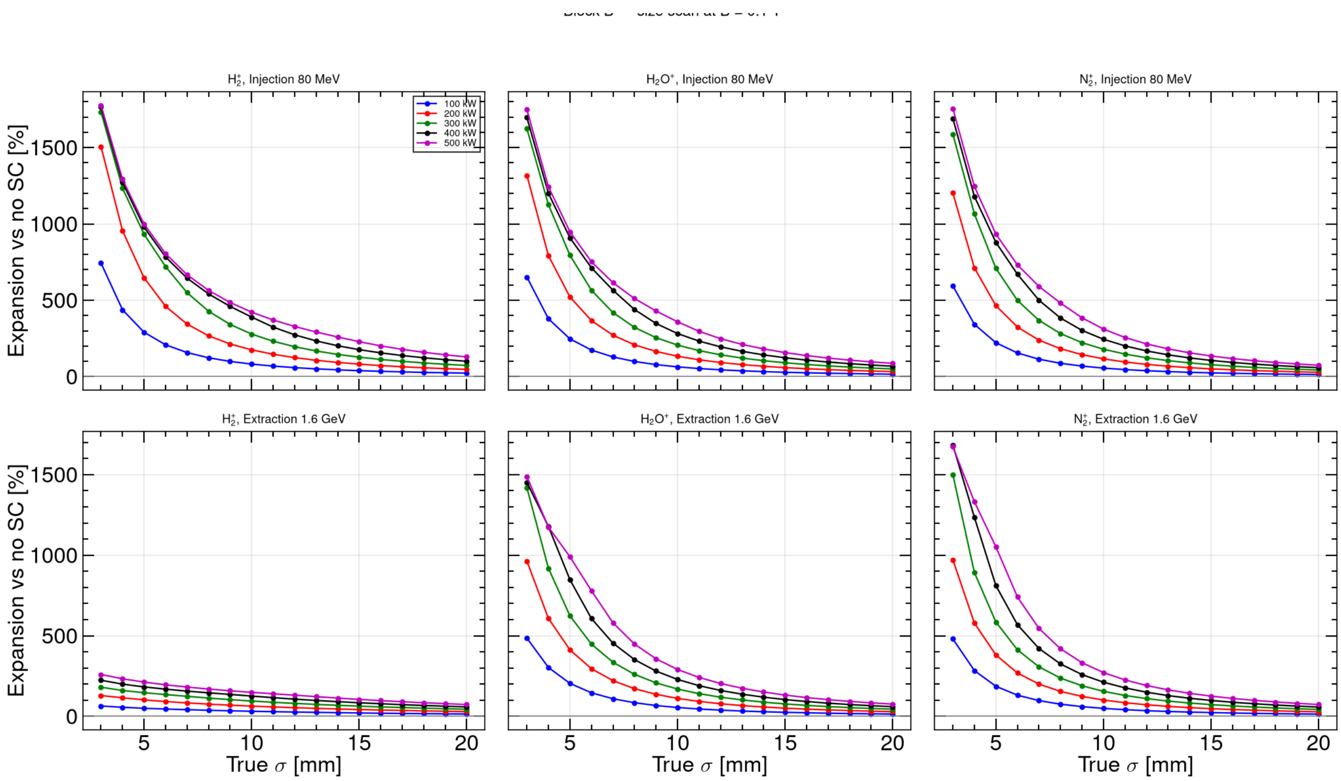

Figure 3. Expansion vs true σ_x at **0.1 T** for 100–500 kW. Smaller beams and higher power give dramatically larger expansion; injection is worse than extraction for H_2^+, but H_2O^+ and N_2^+ can be worse at extraction (longer ToF).

Table 2 — obtained size at σ_x = 10 mm, 0.1 T, 100 kW

Species	Injection		Extraction	
	obtained [mm]	expansion	obtained [mm]	expansion
H_2^+	18.2	+82%	13.2	+32%
H_2O^+	16.4	+64%	15.5	+55%
N_2^+	15.5	+56%	15.0	+50%

Table 3 — H_2^+ injection, σ=10 mm, 0.1 T vs power

100 kW	200 kW	300 kW	400 kW	500 kW
+82% (18.2 mm)	+175% (27.5 mm)	+279% (37.8 mm)	+390% (48.9 mm)	+422% (52.1 mm)

Worst case in the matrix: **injection, σ=3 mm, 500 kW, 0.1 T** — obtained σ ≈ 56 mm (+1700%+) for all three species (cage-wall saturation).

5. Block C — cage voltage (5–30 kV)

Injection 10×10 mm; voltages 5, 10, 15, 20, 25, 30 kV; B = 0, 200, 1000 G; 100 and 500 kW.

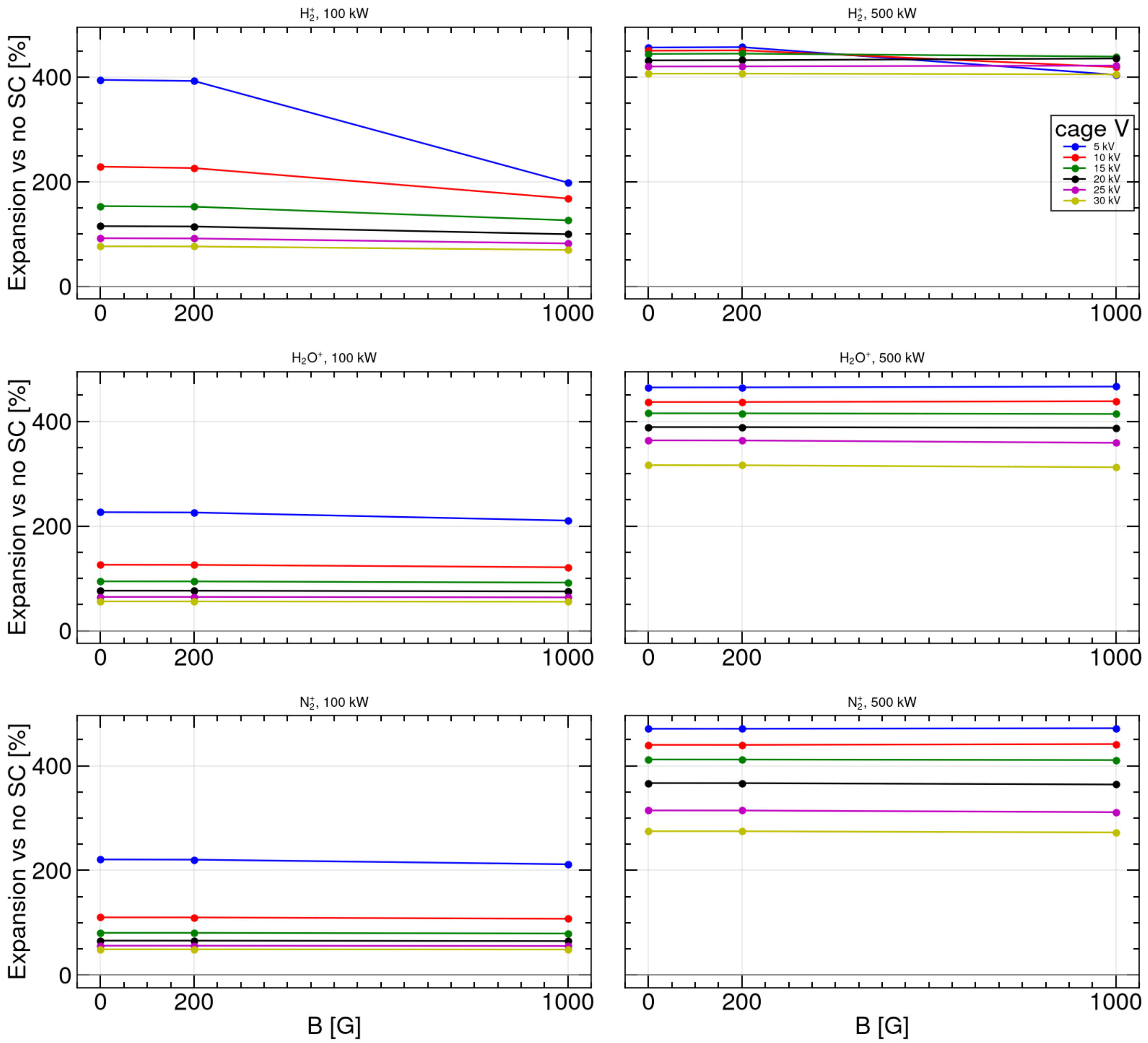

Figure 4. Expansion vs B for each cage voltage. Unlike e-mode Block C, ion expansion **never changes sign** — it stays positive at every V. Higher voltage shortens ion drift time and reduces the integrated space-charge kick.

Table 4 — H_2^+ injection 10 mm, 100 kW, 0.1 T vs cage voltage

5 kV	10 kV	15 kV	20 kV	25 kV	30 kV
+198%	+168%	+126%	+100%	+82%	+70%

At 500 kW the same trend holds but at much higher expansion (+300%–800% range).

6. Block D — beam offset

Injection and extraction 10×10 mm; offsets (+5,0), (+10,0), (0,+5), (0,−5) mm; B = 0, 200, 1000 G; 100 and 500 kW.

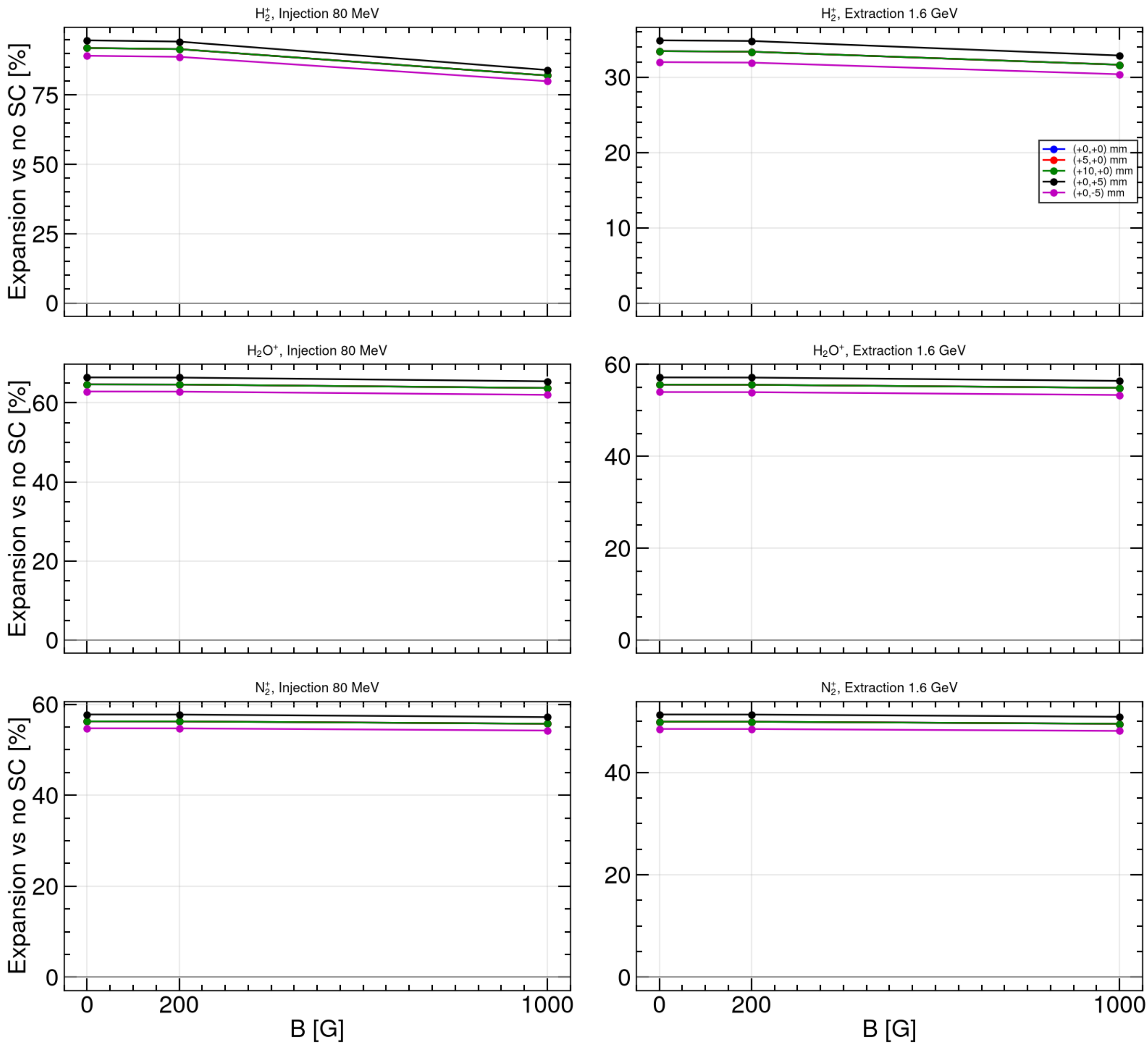

Figure 5. Expansion vs B for each offset (100 kW). Δ x = +5 and +10 mm give **identical expansion** to the centred beam (translation invariance). Δ y = ±5 mm changes expansion by a few percent (toward the detector: slightly less expansion).

Table 5 — H_2^+ injection 10 mm, 0.1 T, 100 kW vs offset

centred	Δ x=+5	Δ x=+10	Δ y=+5	Δ y=-5
+82.1%	+82.1%	+82.1%	+84.0%	+80.0%

Centroid shift vs no-SC is $\lesssim$ 0.6 mm at extraction, 500 kW.

7. Comparison with e-mode (round-beam replan)

	E-mode (§8)	Ion-mode (this report)
Secondaries	electrons	H_2^+ / H_2O^+ / N_2^+
B grid	dense (5 G / 25 G)	0, 200, 1000 G only
Effect of B	oscillatory recovery; 300 G fixes profile	flat; no recovery
Cage voltage	sign flip at low B	always positive; V tunes magnitude
Beam offset	Δ x invariant; Δ y matters	same
Worst case	low B, wrong polarity	small σ, high P, injection

Practical implication: tune the IPM guiding field and cage voltage for **faithful electron profiles**; treat **ion-mode size measurements as space-charge biased** unless beam size and power are large enough (or a correction model is applied).

8. Data products

File	Contents
`output/csns_imode_bscan_summary.csv`	Block A (135 rows)
`output/csns_imode_size_summary.csv`	Block B (1620 rows)
`output/csns_imode_voltage_summary.csv`	Block C (108 rows)
`output/csns_imode_offset_summary.csv`	Block D (180 rows)
`plots/csns_imode_*.png`	Figures 1–5 above
`configs/csns_rcs_ipm/imode/README.md`	Matrix specification

CSV files under output/imode/ (particle lists) are gitignored.